

Numerical Differentiation

$y'(x) \approx P'_n(x)$ (key idea)

Newton's forward difference formula:

$$P_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots$$

$\hookrightarrow (i)$

Differentiating (i) w.r.t x ,

$(ph = x - x_0)$

$\downarrow$

$$y'(x) \approx P'_n(x) = \frac{d}{dx} P_n(x) = \frac{d}{dp} P_n(x) \cdot \frac{dp}{dx}$$

$$\frac{dp}{dx} = \frac{1}{h}$$

$$P'_n(x) = \frac{1}{h} \left[0 + \Delta y_0 + \frac{(2p-1)}{2!}\Delta^2 y_0 + \frac{(3p^2-6p+2)}{3!}\Delta^3 y_0 + \dots \right]$$

$$\left[\left(\frac{dy}{dx} \right)_x = \frac{1}{h} \left[\Delta y_0 + \frac{(2p-1)}{2!}\Delta^2 y_0 + \frac{(3p^2-6p+2)}{3!}\Delta^3 y_0 + \dots \right] \right]$$

At $x = x_0$,

$$p = \frac{x - x_0}{h} = 0$$

$$\left(\frac{dy}{dx} \right)_{x_0} = \frac{1}{h} \left[\Delta y_0 + \frac{(-1)}{2!}\Delta^2 y_0 + \frac{(2)}{3!}\Delta^3 y_0 + \dots \right]$$

$$\left[\left(\frac{dy}{dx} \right)_{x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 + \dots \right] \right] \begin{matrix} \text{Derivative} \\ \rightarrow \text{at initial} \\ \text{pt} \end{matrix}$$

$$\Rightarrow \left(\frac{d^2 y}{dx^2} \right)_{x=x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^2 y_0 + \frac{11}{12}\Delta^4 y_0 - \frac{5}{6}\Delta^5 y_0 + \dots \right]$$

Eg: From the following values of x & y obtained y' for $x=2$.

Ans: - 3.3205

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.0	2.7183	0.6018					
1.2	3.3201	0.7951	0.1833				
1.4	4.0552	0.8948	0.1624	0.0294			
1.6	4.9530	1.0966	0.1988	0.0361	0.0067		
1.8	6.0496	1.3396	0.2429	0.0441	0.0080	0.0013	
2.0	7.3891	1.6359	0.2964	0.0535	0.0094	0.0009	
2.2	9.0250						

$$y' = \frac{1}{h} \left[\Delta y_0 + \frac{(2p-1)}{2!} \Delta^2 y_0 + \frac{(3p^2-6p+2)}{3!} \Delta^3 y_0 + \dots \right]$$

$$h = 0.2$$

$$p = \frac{x - x_0}{h} = \frac{1.2 - 1.0}{0.2} = \frac{0.2}{0.2} = 1$$

$$\begin{aligned}
 y' = \frac{1}{h} & \Delta y_0 + \frac{(2p-1)}{2!} \Delta^2 y_0 + \frac{(3p^2-6p+2)}{3!} \Delta^3 y_0 \\
 & + \frac{(4p^3-18p^2+22p-6)}{4!} \Delta^4 y_0 \\
 & + \frac{(5p^4-40p^3+105p^2-100p+24)}{5!} \Delta^5 y_0 \\
 & + \frac{(6p^5-75p^4+340p^3-675p^2+548p-120)}{6!} \Delta^6 y_0
 \end{aligned}$$

$$= 8.92028$$

$$\textcircled{2} \int_{x_0}^{x_2}$$

$$\textcircled{3} \int_{x_0}^{x_2}$$

$$\int_a^b f(x) dx$$

$$\int_a^b f(x) dx$$

Eg. For

$$h = 0.2$$

$$x_0 = 0$$

Numerical Integration :-

$\int_0^1 \sin^2 x \, dx$ - non-integrable form.

$$\int_a^b f(x) \, dx = \int_a^b P_n(x) \, dx \quad (f(x) \approx P_n(x))$$

$$f(x) = P_n(x) + \text{error} \quad (n=1)$$

$$\int_{a=x_0}^{b=x_1} f(x) \, dx = \int_{x_0}^{x_1} \left[P_n(x) + \frac{(x-x_0)(x-x_1)}{2!} f''(\xi) \right] dx; \quad x_0 < \xi < x_1$$

$$= \underbrace{\int_{x_0}^{x_1} P_n(x) \, dx}_{(i)} + \underbrace{\int_{x_0}^{x_1} \frac{(x-x_0)(x-x_1)}{2!} f''(\xi) \, dx}_{(ii) \text{ Error}}$$

$$(i) \int_{x_0}^{x_1} P_1(x) \, dx = \int_{x_0}^{x_1} [L_0(x)f(x_0) + L_1(x)f(x_1)] \, dx$$

$$= f(x_0) \int_{x_0}^{x_1} \frac{(x-x_1)}{(x_0-x_1)} \, dx + f(x_1) \int_{x_0}^{x_1} \frac{(x-x_0)}{(x_1-x_0)} \, dx$$

$$= \left. \frac{f(x_0)}{(x_0-x_1)} \frac{(x-x_1)^2}{2} \right|_{x_0}^{x_1} + \left. \frac{f(x_1)}{(x_1-x_0)} \frac{(x-x_0)^2}{2} \right|_{x_0}^{x_1}$$

$$= \frac{f(x_0)}{(x_0-x_1)} \left(-\frac{(x_0-x_1)^2}{2} \right) + \frac{f(x_1)}{(x_1-x_0)} \frac{(x_1-x_0)^2}{2}$$

$$= \frac{(x_1-x_0)^2}{2} \frac{[f(x_0) + f(x_1)]}{(x_1-x_0)}$$

$$= \frac{(x_1-x_0)}{2} [f(x_0) + f(x_1)]$$

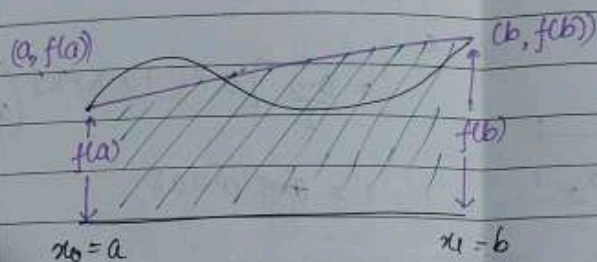
= Area of trapezoidal

Trapezoidal law:

$$\int_{x_0=a}^{x_1=b} f(x) dx = \int_{x_0=a}^{x_1=b} p(x) dx = \frac{(x_1 - x_0)}{2} [f(x_0) + f(x_1)]$$

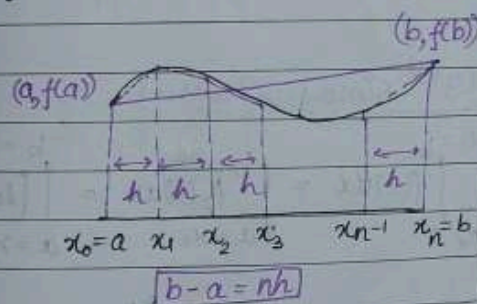
$$= \left(\frac{b-a}{2} \right) [f(a) + f(b)]$$

= Area of trapezoidal.



Complete trapezoidal law: dividing the interval into multiple trapezoidal.

$$\int_{x_0=a}^{x_n=b} f(x) dx = \int_{x_0}^{x_1} f(x) dx + \int_{x_1}^{x_2} f(x) dx + \dots + \int_{x_{n-1}}^{x_n} f(x) dx$$



$$= \frac{h}{2} [f(x_0) + f(x_1)] + \frac{h}{2} [f(x_1) + f(x_2)] + \frac{h}{2} [f(x_2) + f(x_3)] + \dots + \frac{h}{2} [f(x_{n-1}) + f(x_n)]$$

$$= \frac{h}{2} [f(x_0) + 2 \{f(x_1) + f(x_2) + \dots + f(x_{n-1})\} + f(x_n)]$$

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$$= \frac{(x_1 - x_0)}{2} [f(x_0) + f(x_1)]$$

= Area of trapezoidal

eg. For the function $f(x) = \frac{1}{x+1}$, find $\int_0^1 f(x) dx$ using trapezoidal law & complete trapezoidal law with $h=0.5$. Also find Error.

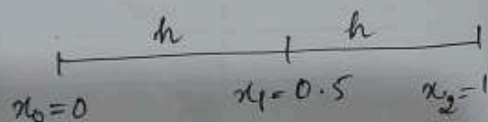
$$\int_{a=x_0}^{b=x_2} f(x) dx = \left(\frac{b-a}{2}\right)(f(b)+f(a)) = \left(\frac{1-0}{2}\right)(f(1)+f(0)) = \frac{1}{2} \left(\frac{1}{2}+1\right) = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4} = 0.75$$

$$\text{Error} = \left| \ln(x) \Big|_0^1 - 0.75 \right| = \left| [\ln(2) - \ln(1)] - 0.75 \right| = 0.05$$

Complete trapezoidal,

$$nh = b-a$$

$$n = \frac{1-0}{0.5} = 2$$



$$\begin{aligned} \int_0^1 \frac{1}{x+1} dx &= \frac{h}{2} \{ f(x_0) + f(x_2) + 2f(x_1) \} \\ &= \frac{0.5}{2} [1 + 1 + 2 \cdot 2] = \frac{0.5}{2} [7] = 0.708 \end{aligned}$$

$$E_a = \left| (\ln(2) - \ln(1)) - 0.708 \right| = 0.01485$$

1/3rd Simpson's law:-

$$\int_{a=x_0}^{b=x_2} f(x) dx = \int_{a=x_0}^{b=x_2} p_2(x) dx = \int_{a=x_0}^{b=x_2} [l_0(x)f(x_0) + l_1(x)f(x_1) + l_2(x)f(x_2)] dx$$

$$= f(x_0) \int_{x_0}^{x_2} l_0(x) dx + f(x_1) \int_{x_0}^{x_2} l_1(x) dx + f(x_2) \int_{x_0}^{x_2} l_2(x) dx$$

(1) (2) (3)

$$= 3.32029$$

trapezoidal law
of Exakte.

$$+1) = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$$

$$= 0.75$$

$$f(x_0) + f(x_1) + 2f(x_2) + f(x_3)$$

$$+1 + 0.5 \times 2 + \left(\frac{1}{0.1} + \frac{1}{1.1} + \frac{2}{1.05} \right)$$

$$0.708$$

$$(x_1) + \int_{x_0}^{x_2} f(x) dx$$

$$+ f(x_2) \int_{x_0}^{x_2} f(x) dx$$

③

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$$\text{Now, } \int_{x_0=a}^{x_1=b} f_2(x) dx = \int_a^b \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} dx$$

$$= \int_a^b \frac{(x - \frac{a+b}{2})(x-b)}{(\frac{a-b}{2})(a-b)} dx$$

$$= \frac{2}{(a-b)^2} \int_a^b (x - \frac{a+b}{2})(x-b) dx$$

$$= \frac{2}{(a-b)^2} \int_{t=-1}^{t=+1} (\frac{b-a}{2})^2 t(t-1) (\frac{b-a}{2}) dt$$

$$= \frac{2}{(a-b)^2} \cdot \frac{(b-a)^2}{2^2} \cdot \frac{(b-a)}{2} \left[\frac{t^3}{3} - \frac{t^2}{2} \right]_{-1}^{+1}$$

$$= \frac{2}{(a-b)^2} \cdot \frac{(b-a)^2}{2^2} \cdot \frac{(b-a)}{2} \cdot \left[-\frac{1}{3} - \frac{1}{2} - \left(-\frac{1}{3} + \frac{1}{2} \right) \right]$$

$$= \frac{(b-a)}{2^2} \cdot \frac{+2}{3}$$

$$= \frac{+(b-a)}{3 \cdot 2} = \frac{+(b-a)}{6}$$

$$\text{Let } x = \frac{(b+a)}{2} + \left(\frac{b-a}{2} \right) t$$

$$dx = \left(\frac{b-a}{2} \right) dt$$

Limit:

$$x=a; \quad t=-1$$

$$x=b; \quad t=1$$

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$$\textcircled{2} \int_{x_0=a}^{x_2=b} L_1(x) dx = \left(\frac{1}{6}\right)(b-a)$$

$$\textcircled{3} \int_{x_0=a}^{x_2=b} L_2(x) dx = \left(\frac{b-a}{6}\right)$$

Generalized

$$\int_a^b f(x) dx = \left(\frac{b-a}{6}\right) \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$= \frac{2h}{6} \left\{ f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right\}$$

$$\begin{cases} b-a = nh \\ \downarrow \\ n = 2 \\ \text{degree} \end{cases}$$

$$\int_a^b f(x) dx = \frac{h}{3} \left[f(x_0) + 4f(x_1) + f(x_2) \right]$$

$$x_0 = a \quad x_1 = \frac{a+b}{2} \quad x_2 = b$$

Ex: For the function $f(x) = e^{-x^2}$. Find $\int_0^1 f(x) dx$ using $\frac{1}{3}$ rd Simpson's law

$$h = 0.5$$

$$x_0 = 0 \quad x_1 = 0.5 \quad x_2 = 1$$

$$\int_0^1 e^{-x^2} dx = \frac{0.5}{3} \left\{ f(0) + 4f(0.5) + f(1) \right\}$$

$$= \frac{0.5}{3} \left\{ e^{-0} + 4e^{-0.5^2} + e^{-1} \right\}$$

$$= 0.74418$$

Numerical Integra

$\int_0^1 \sin^2 x dx$ - non-inter

$$\int_a^b f(x) dx = \left[f(x) \right]_a^b$$

$$f(x) = P_n(x) + \text{error}$$

$$\int_a^b f(x) dx = \int_a^b P_n(x) dx + \text{error}$$

$$= \int_a^b P_n(x) dx$$

$$(i) \int_{x_0}^{x_1} P_1(x) dx = \int_{x_0}^{x_1} f(x) dx$$

Composite Vs Simpson's law:

$$a=x_0 \quad x_1 \quad x_2 \quad x_3 \quad x_4 \quad \dots \quad x_{n-1} \quad x_n=b$$

Simpson's law:

$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] ; \text{ where } x_i = x_0 + ih$$

$i = 0, 1, 2, 3, \dots, n$

Composite Simpson law:-

$n \rightarrow \text{even}$

$$\int_{a=x_0}^{b=x_n} f(x) dx = \sum_{j=1}^{n/2} \int_{x_{2j-2}}^{x_{2j}} f(x) dx$$

$$= \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \dots + \int_{x_{2j-2}}^{x_{2j}} f(x) dx$$

$$+ \dots + \int_{x_{n-2}}^{x_n} f(x) dx$$

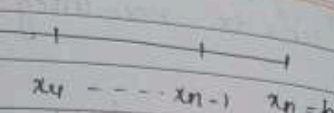
$$= \sum_{j=1}^{n/2} \left(\frac{h}{3} \right) [f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j})]$$

$$= \left(\frac{h}{3} \right) [f(x_0) + 4f(x_1) + f(x_2)] + \left(\frac{h}{3} \right) [f(x_2) + 4f(x_3) + f(x_4)]$$

$$+ \left(\frac{h}{3} \right) [f(x_4) + 4f(x_5) + f(x_6)] + \dots$$

$$+ \left(\frac{h}{3} \right) [f(x_{n-4}) + 4f(x_{n-3}) + f(x_{n-2})]$$

$$+ \left(\frac{h}{3} \right) [f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$$



(x_i) ; where $x_i = x_0 + i h$
 $i = 0, 1, 2, 3, \dots$

$$\int_{x_0}^{x_n} f(x) dx = \int_{x_0}^{x_1} f(x) dx + \dots + \int_{x_{n-1}}^{x_n} f(x) dx$$

$$= h \left[\frac{f(x_0) + f(x_n)}{2} + f(x_1) + 2f(x_2) + \dots + 2f(x_{n-2}) + f(x_{n-1}) \right]$$

$$\int_{a=x_0}^{b=x_n} f(x) dx = \frac{h}{3} \left[f(x_0) + 4[f(x_1) + f(x_3) + \dots + f(x_{n-1})] + 2[f(x_2) + f(x_4) + \dots + f(x_{n-2})] + f(x_n) \right]$$

Find the area bounded by the curve and the x-axis from $x = 1.47$ to 1.53 using following table:

x	$f(x)$
$x_0 = 1.47$	$f(x_0) = 1.93$
$x_1 = 1.48$	$f(x_1) = 1.95$
$x_2 = 1.49$	$f(x_2) = 1.99$
$x_3 = 1.50$	$f(x_3) = 2.01$
$x_4 = 1.51$	$f(x_4) = 2.03$
$x_5 = 1.52$	$f(x_5) = 2.03$
$x_6 = 1.53$	$f(x_6) = 2.08$

by trapezoidal & Simpson's formula.

Simpson

$$\Rightarrow \text{Area} = \int_{x=1.47}^{x=1.53} f(x) dx = \frac{h}{3} \left[f(x_0) + f(x_6) + 4[f(x_1) + f(x_3) + f(x_5)] + 2[f(x_2) + f(x_4)] \right]$$

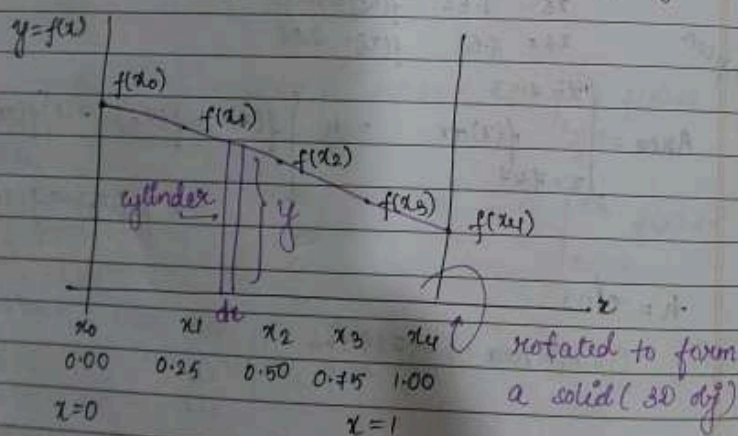
$h = 0.01$
 Area = 0.119966

$\Rightarrow$ Trapezoidal: $\frac{h}{2} [f(x_0) + 2\{f(x_1) + \dots + f(x_5)\} + f(x_6)]$
 $= 0.12005$

eg A solid of revolution is formed by rotating about the x-axis, the area b/w the x-axis, the lines $x=0$ and $x=1$, and a curve through the points with the following coordinates.

x	$y=f(x)$
0.00 x_0	1.0000
0.25 x_1	0.9896
0.50 x_2	0.9589
0.75 x_3	0.9089
1.00 x_4	0.8915

Find the volume of solid formed by overlapping.



$$\begin{aligned}
 V &= \int_0^1 \pi y^2 dx \quad (V = \pi r^2 h) \\
 &= \pi \int_0^1 y^2 dx \quad (y = y^2) \\
 &= \pi \frac{1}{3} \left[y_0^3 + 4(y_1^3 + y_2^3) + 2y_3^3 + y_4^3 \right] = 2.84193
 \end{aligned}$$

Degree of Accuracy
The degree of precision/accuracy (eg Simpson's rule) is the number of points used for each interval h such that the error is $O(h^k)$ for each h .

For Simpson's formula
 $\int_a^b f(x) dx = \frac{b-a}{2} [f(a) + f(b)]$

Analytical value

$$(a) \int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

$$(b) f(x) = x \quad \int_0^1 x dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}$$

$$(c) f(x) = x^2 \quad \int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

For Simpson's formula

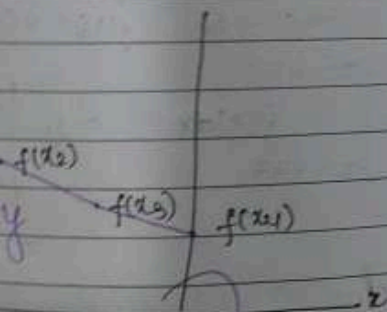
$$\int_a^b f(x) dx = \frac{b-a}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)]$$

formed by rotating about the axis, the line joining the points with

$$y=f(x)$$

1.0000
0.9896
0.9589
0.9089
0.8415

id formed by overlapping.



rotated to form a solid (3D obj)

$x=1$

$$(V = \pi r^2 h)$$

$$(V = \pi r^2 h)$$

$$+4(y_1 + y_2) + 2y_3 + y_4 = 2.84193$$

Degree of Precision :-

The degree of precision/accuracy of a quadrature formula (eg. trapezoidal, Simpson's formula etc) is the largest positive integer n such that the formula is true (exact) for x^k , for each $k=0,1,2,\dots,n$ i.e. $x^0, x^1, x^2, \dots, x^n$.

For trapezoidal formula:

$$\int_a^b f(x) dx = \left(\frac{b-a}{2}\right) [f(a) + f(b)]$$

Analytical value -

$$(a) \int_0^1 f(x) dx = \int_0^1 x^0 dx = (x^1)' = 1$$

Numerical value by formula -

$$= \int_0^1 x^0 dx = \left(\frac{1-0}{2}\right) [1+1] = 1$$

$$(b) f(x)=x$$

$$\int_0^1 x dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$$

$$= \int_0^1 x dx = \left(\frac{1-0}{2}\right) (f(1) + f(0))$$

$$= \frac{1}{2} (1+0) = \frac{1}{2}$$

$$(c) f(x)=x^2$$

$$\int_0^1 x^2 dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$$

$$\neq \int_0^1 x^2 dx = \frac{1}{2} [f(1) - f(0)]$$

$$= \frac{1}{2} (1-0) = \frac{1}{2}$$

DOP of trapezoidal = $K=1$

For Simpson's formula:-

$$\int_a^b f(x) dx = \left(\frac{b-a}{6}\right) [f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)]$$

Analytical

$$(a) \int_0^1 x^0 dx = (x)_0^1 = 1$$

Numerical

$$= \int_0^1 x^0 dx = \left(\frac{1-0}{6}\right) [f(1) + 4f(0.5) + f(0)]$$

$$= \frac{1}{6} (1 + 4 + 1) = 1$$

$$(b) f(x) = x^1$$

$$\int_0^1 x^1 dx = \left(\frac{x^2}{2}\right)_0^1 = \frac{1}{2}$$

$$= \int_0^1 x^1 dx = \frac{1}{6} [f(1) + 4f(0.5) + f(0)]$$

$$= \frac{1}{6} (1 + 2 + 0) = \frac{1}{2}$$

Order of Simpson = 2

$$(c) f(x) = x^2$$

$$\int_0^1 x^2 dx = \left(\frac{x^3}{3}\right)_0^1 = \frac{1}{3}$$

$$= \int_0^1 x^2 dx = \frac{1}{6} [f(1) + 4f(0.5) + f(0)]$$

$$= \frac{1}{6} (1 + 1 + 0) = \frac{1}{3}$$

$$(d) f(x) = x^3$$

$$\int_0^1 x^3 dx = \left(\frac{x^4}{4}\right)_0^1 = \frac{1}{4}$$

$$= \int_0^1 x^3 dx = \frac{1}{6} \left(1 + \frac{1}{2} + 0\right) = \frac{1}{6} \times \frac{3}{2} = \frac{1}{4}$$

$$\frac{1}{6} \times \left(1 + \frac{1}{2}\right) = \frac{1}{6} \times \frac{3}{2} = \frac{1}{4}$$

$$(e) f(x) = x^4$$

$$\int_0^1 x^4 dx = \left(\frac{x^5}{5}\right)_0^1 = \frac{1}{5}$$

Order K=3

$$\# \text{ For Simpson's formula}$$

$$\frac{1}{6} (f(0) + 4f(0.5) + f(1))$$

More is the Order, More is the accuracy

$$\frac{1}{6} (0 + 1 + 4 \times \frac{625}{16})$$

$$= \frac{1}{6} (0 + 1 + 156.25)$$

$$= \frac{1}{6} \times \frac{157.25}{1} = \frac{157.25}{6}$$

② calc. using simp

$$b-a = nh$$

$$\frac{\pi}{2} - 0 = 6 \times h$$

$$h = \frac{\pi}{12}$$

$$\int_0^{\pi/2} \sin x dx = \left(-\cos x\right)_0^{\pi/2} = \frac{1}{2}$$

② Use the trap integral.

$$b-a = nh$$

$$2-0 = 6h$$

$$h = \frac{1}{3}$$

$$f(x) = \frac{1}{1+x^6}$$

$$= \frac{1}{3} \times \frac{1}{2} \left[\frac{1}{1+0^6} + \frac{1}{1+1^6} \right]$$

$$= \frac{1}{3} \times \frac{1}{2} \left[1 + \frac{1}{2} \right]$$

$$= \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$$

Numerical

$$\int_0^1 x^2 dx = \frac{(1-0)}{6} [f(1) + 4f(0.5) + f(0)]$$

$$= \frac{1}{6} (1 + 4 + 1) = 1$$

$$\int_0^1 x^2 dx = \frac{1}{6} [f(1) + 4f(0.5) + f(0)]$$

$$= \frac{1}{6} (1 + 4 + 1) = \frac{7}{6}$$

$$f(x) = x^2$$

$$\frac{1}{6} [f(1) + 4f(0.5) + f(0)]$$

$$\frac{1}{6} [1 + 1 + 0] = \frac{1}{3}$$

$$\frac{1}{6} (1 + \frac{4}{3} + 0) = \frac{1}{6} \times \frac{7}{3} = \frac{7}{18}$$

$$\frac{1}{6} \times (1 + \frac{4}{3}) = \frac{1}{6} \times \frac{7}{3} = \frac{7}{18}$$

For Simpson's formula

$$\frac{1}{6} [f(0) + 4f(0.5) + f(1)]$$

the accuracy

$$\frac{1}{6} [0 + 1 + 4 \times \frac{7}{18}]$$

$$= \frac{1}{6} [0 + 1 + \frac{28}{9}]$$

$$= \frac{1}{6} \times \frac{37}{9} = \frac{37}{54}$$

Tutorial-

① calc. using Simpson's 1/3 rule $\int_0^{\pi/2} \sin x dx, n=6$

$$b-a = nh$$

$$\frac{\pi}{2} - 0 = 6 \times h$$

$$h = \pi/12$$

$$\int_0^{\pi/2} \sin x dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

x_0	x_1	x_2	x_3	x_4	x_5	x_6
0	$\frac{\pi}{12}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{5\pi}{12}$	$\frac{\pi}{2}$

$$\int_0^{\pi/2} \sin x dx = \frac{h}{3} [f(0) + 4(f(\pi/12) + f(\pi/4) + f(5\pi/12)) + 2(f(\pi/6) + f(\pi/3)) + f(\pi/2)]$$

$$= 1.18728$$

② Use the trapezoidal formula to approximate the given integral. $\int_0^2 \frac{1}{1+x^6} dx; n=6$

$$= \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + 2f(x_4) + 2f(x_5) + f(x_6)]$$

$$b-a = nh$$

$$2-0 = 6h$$

$$h = \frac{1}{3}$$

y_0	y_1	y_2	y_3	y_4	y_5	y_6
1	0.9986	0.9193	0.5	0.151	0.0446	0.01538
$x_0=0$	$x_1=\frac{1}{3}$	$x_2=\frac{2}{3}$	$x_3=1$	$x_4=\frac{4}{3}$	$x_5=\frac{5}{3}$	$x_6=2$

$$f(x) = \frac{1}{1+x^6}$$

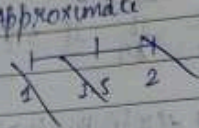
$$= \frac{1}{2} \times \frac{1}{3} [f(0) + 2f(\frac{1}{3}) + 2f(\frac{2}{3}) + 2f(1) + 2f(\frac{4}{3}) + 2f(\frac{5}{3}) + f(2)]$$

$$= 1.0404$$

eg. use the trapezoidal rule to compute the given integral using 3 intervals. Compare it to exact value $\int_1^2 \frac{dx}{x}$ ($n=3$)

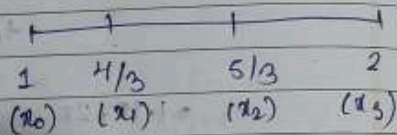
Actual: $\int_1^2 \frac{dx}{x} = \ln x \Big|_1^2 = 0.693$

Approximate



$h=0.5$

$h = \frac{2-1}{3} = \frac{1}{3}$ ($n=3$)



$$\Rightarrow \frac{h}{2} [f(x_0) + 2\{f(x_1) + f(x_2)\} + f(x_3)]$$

 $= 0.7$

$E_a = 0.007$

% Er = 0.11

eg. Find the degree of precision of the following approximations for $\int_{-1}^1 f(x) dx$

actual

(a) $f(x) = x^0$

$\int_{-1}^1 dx = 2$

(ii) $\frac{2}{3} [f(-1) + f(0) + f(1)]$

(iii) $f(-1/\sqrt{3}) + f(1/\sqrt{3})$ approximate

$f(x) = 1$

$\int_{-1}^1 x^0 dx = \left(\frac{1-(-1)}{2}\right) [f(0) + f(1)]$

$= 1(1+1) = 2$

(b) $f(x) = x^1$

$\int_{-1}^1 x dx = \frac{x^2}{2} \Big|_{-1}^1 = 0$

$\Rightarrow \frac{2}{2} (1-1) = 0$

(c) $f(x) = x^2$
 $\int_{-1}^1 x^2 dx = \frac{x^3}{3} \Big|_{-1}^1 = \frac{1+1}{3}$
 $[80p = K = 1]$

(K) $\frac{2}{3} [f(-1) + f(0) + f(1)]$

Actual

(a) $f(x) = x^0 = 1$

$\int_{-1}^1 dx = 2$

(b) $f(x) = x$

$\int_{-1}^1 x dx = 0$

(c) $f(x) = x^2$

$\int_{-1}^1 x^2 dx = 2/3$

(iii) approx

$f(-1/\sqrt{3}) + f(1/\sqrt{3})$

$f(x) = 1$

$f(x) = x$

$f(x) = x^2$

$f(x) = x^3$

$f(x) = x^4$

$[K = 3]$

compute the given integral
 take it exact value

0.693

(c) $f(x) = x^2$
 $\int_{-1}^1 x^2 dx = \left. \frac{x^3}{3} \right|_{-1}^1 = \frac{1+1}{3} = \frac{2}{3} \neq \frac{2}{3}(1+1) = 2$
loop = K=1

(iv) $\frac{2}{3} [f(-1) + f(0) + f(1)]$

Approximate

(a) $f(x) = x^0 = 1$
 $\int_{-1}^1 dx = 2$

$\frac{2}{3} [1+1+1] = 2$

(b) $f(x) = x$
 $\int_{-1}^1 x dx = 0$

$\frac{2}{3} [-1+0+1] = 0$

(c) $f(x) = x^2$
 $\int_{-1}^1 x^2 dx = \frac{2}{3}$

$\frac{2}{3} [1+0+1] = \frac{4}{3} \neq \frac{2}{3}$

K=1

(iii) approx

$\frac{1}{3} [f(-1/3) + f(1/3)]$

approx actual

$f(x) = 1$

$1+1 = 2 = 2$

$f(x) = x$

$0 = 0$

$f(x) = x^2$

$\frac{1}{3} \left(\frac{1}{9} + \frac{1}{9} \right) = \frac{2}{9} = \frac{2}{9}$

$f(x) = x^3$

$\frac{-1}{3\sqrt{3}} + \frac{1}{3\sqrt{3}} = 0 = 0$

$\int_{-1}^1 x^3 dx = \left. \frac{x^4}{4} \right|_{-1}^1 = 0$

$f(x) = x^4$

$\frac{2}{9}$

$\int_{-1}^1 x^4 dx = \left. \frac{x^5}{5} \right|_{-1}^1 = \frac{2}{5}$

K=2

of the following approximations

(i) $\frac{2}{3} [f(-1) + f(0) + f(1)]$

$\frac{2}{3} [f(-1/3) + f(1/3)]$ approximate

$f(x) = 1$

$\int_{-1}^1 x^0 dx = \left(\frac{1-(-1)}{2} \right) [f(0) + f(1)]$

$= 1(1+1) = 2$

$\frac{2}{3} (1-1) = 0$

Numerical Integral formula

eg the quadrature formula: $\int_{-1}^1 f(x) dx = C_0 f(-1) + C_1 f(0) + C_2 f(1)$

is exact for all polynomials of degree less than or equal to 2. Determine C_0, C_1, C_2

DOP = 2

$$f(x) = \{x^0, x^1, x^2\}$$

Actual

$$f(x) = x^0 = 1 \quad \int_{-1}^1 dx = 2$$

$$f(x) = x^1 \quad \left. \frac{x^2}{2} \right|_{-1}^1 = 0$$

$$f(x) = x^2 \quad \left. \frac{x^3}{3} \right|_{-1}^1 = \frac{2}{3}$$

Approximate

$$C_0 + C_1 + C_2 = 2$$

$$-C_0 + C_2 = 0$$

$$C_0 + C_2 = \frac{2}{3}$$

$$C_2 = \frac{1}{3} = C_0$$

$$C_1 = 2 - \frac{2}{3} = \frac{4}{3} = C_1$$

$$\text{eg } \int_0^1 f(x) dx = C_0 f(0) + C_1 f(0.5) + C_2 f(1) \quad ; \text{DOP} > 1$$

actual

$$f(x) = x^0 = 1 \quad \int_0^1 dx = 1$$

$$f(x) = x \quad \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}$$

$$f(x) = x^2 \quad \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}$$

approx.

$$C_0 + C_1 + C_2 = 1$$

$$\frac{1}{2} C_1 + C_2 = \frac{1}{2} \Rightarrow 1 = C_1 + 2C_2$$

$$\frac{1}{4} C_1 + C_2 = \frac{1}{3} \Rightarrow 4 = 3C_1 + 12C_2$$

$$C_1 = \frac{4}{3}$$

$$C_2 = 1 - \frac{C_1}{2} = 1 - \frac{4}{6} = \frac{1}{3}$$

$$C_0 + C_1 + C_2 = 1$$

$$1 = C_1 + 2C_2$$

$$4 = 3C_1 + 12C_2$$

$$3 = 3C_1 + 6C_2$$

$$4 = 3C_1 + 12C_2$$

$$1 = 6C_2$$

$$C_2 = \frac{1}{6}$$

eg Evaluate $\int_0^2 e^x dx$

and $h = 1/2$, compare

$$\text{exact} = \int_0^2 e^x dx$$

Approximate: - (h)

$$\int_0^2 e^x dx = \frac{h}{3} [f(0) + 4f(1) + f(2)]$$

$$= 6.42$$

(h=1/2)

$$\Rightarrow 7.2972$$

$$C_2 = 1 - \frac{C_1}{2} = 1 - \frac{1/3}{2} = 1 - \frac{1}{6} = \frac{5}{6}$$

$$C_2 = 1 - C_1 - C_2 = 1 - \frac{1}{3} - \frac{1}{6} = \frac{1}{2}$$

$$C_2 = 1 - \frac{1}{3} = \frac{2}{3}$$

$$(C_0, C_1, C_2) = (1/3, 1/3, 1/3)$$

$$C_1 = 1 - 2C_2 = 1 - \frac{2}{3} = \frac{1}{3}$$

$$3 \times (ii) \quad 3 = 3C_1 + 6C_2$$

$$1 \times (iii) \quad 4 = 3C_1 + 12C_2$$

$$C_1 = 2/3$$

$$C_2 = 1 - \frac{2}{3} - \frac{1}{3} = \frac{0}{3} = 0$$

$$C_0 = 1/6$$

eg Evaluate $\int_0^2 e^x dx$, using the Simpson's rule with $h=1$

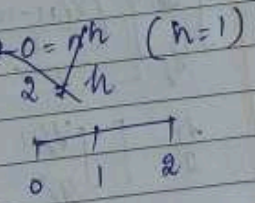
and $h=1/2$, compare with the exact sol?

$$\text{Exact} = \int_0^2 e^x dx = e^x \Big|_0^2 = e^2 - 1 = 6.389$$

Approximate: - ($h=1$)

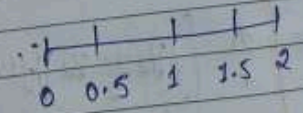
$$\int_0^2 e^x dx = \frac{h}{3} [f(0) + f(2) + 4f(1)]$$

$$= \frac{1}{3} [1 + e^2 + 4e] = 6.42$$

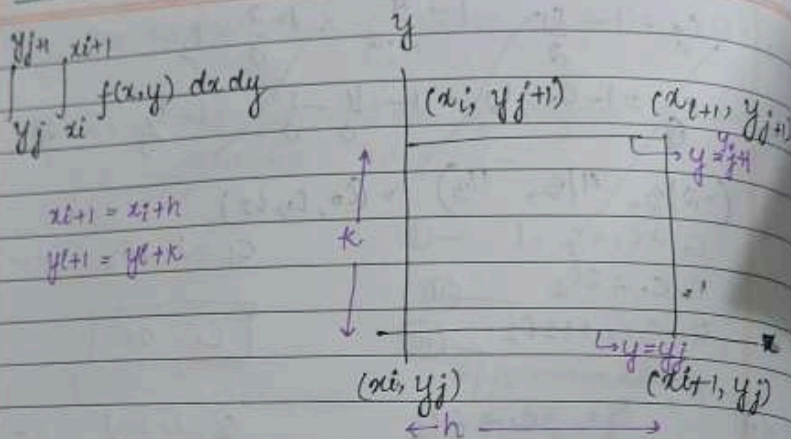


($h=1/2$)

$$\Rightarrow 6.3912$$



Double Integral :-



$$x_{i+1} = x_i + h$$

$$y_{j+1} = y_j + k$$

Trapezoidal:

$$\int_a^b f(x) dx = \frac{(b-a)}{2} [f(b) + f(a)]$$

$$\int_{y_j}^{y_{j+1}} \left(\int_{x_i}^{x_{i+1}} f(x, y) dx \right) dy$$

Trapezoidal Rule

$$= \int_{y_j}^{y_{j+1}} \left[\frac{(x_{i+1} - x_i)}{2} \{ f(x_{i+1}, y) + f(x_i, y) \} \right] dy$$

$$= \frac{h}{2} \left[\int_{y_j}^{y_{j+1}} f(x_i, y) + f(x_{i+1}, y) dy \right]$$

$$= \frac{h}{2} \left[\underbrace{(y_{j+1} - y_j)}_k \{ f(x_i, y_{j+1}) + f(x_i, y_j) \} + \underbrace{(y_{j+1} - y_j)}_k \{ f(x_{i+1}, y_{j+1}) + f(x_{i+1}, y_j) \} \right]$$

$$= \frac{hk}{4} [f(x_i, y_{j+1}) + f(x_i, y_j) + f(x_{i+1}, y_{j+1}) + f(x_{i+1}, y_j)]$$

eg Evaluate

$$= \int_0^1 \left\{ \left(\frac{1-0}{2} \right) (e^{0+y} + e^{1+y}) \right\} dy$$

$$= \frac{1}{2} \int_0^1 (e^y + e^{1+y}) dy$$

$$= \frac{1}{2} \left[\left(\int_0^1 e^y dy \right) + \left(\int_0^1 e^{1+y} dy \right) \right]$$

$$= \frac{1}{2} \left[\left(\frac{1-0}{2} \right) (e^0 + e^1) \right] + \left(\frac{1-0}{2} \right) (e^{0+1} + e^{1+1}) \right]$$

$$= \frac{1}{4} [1 + e + e + e^2]$$

$$h=0.5, k=1$$

$$if (h=k=0.5):$$

$$\int_0^1 \int_0^1 (e^x + y) dx dy$$

$$= \int_0^1 \left\{ \left(\frac{0.5}{2} \right) (e^{0+y} + e^{0.5+y}) \right\} dy$$

$$= \left(\frac{0.5}{2} \right) \left[\left(\int_0^1 e^y dy \right) + \left(\int_0^1 e^{0.5+y} dy \right) \right]$$

$$= \left(\frac{0.5}{2} \right) \left[\left(\frac{1-0}{2} \right) (e^0 + e^1) + \left(\frac{1-0}{2} \right) (e^{0.5+0} + e^{0.5+1}) \right]$$

eg Evaluate: $\int_0^1 \int_0^1 e^{x+y} dx dy$ by using trapezoidal rule for $h=k=1$

$$= \int_0^1 \left[\left(\frac{1-0}{2} \right) (e^{0+y} + e^{1+y}) \right] dy$$

$$= \frac{1}{2} \int_0^1 (e^y + e^{1+y}) dy$$

$$= \frac{1}{2} \left[\left(\int_0^1 e^y dy \right) + \left(\int_0^1 e^{1+y} dy \right) \right]$$

$$= \frac{1}{2} \left[\left(\frac{1-0}{2} \right) (e^0 + e^1) + \left(\frac{1-0}{2} \right) (e^{0+1} + e^{1+1}) \right]$$

$$= \frac{1}{4} [1 + e + e + e^2] = \frac{1+2e+e^2}{4} = 3.4564$$

$h=0.5, k=1$

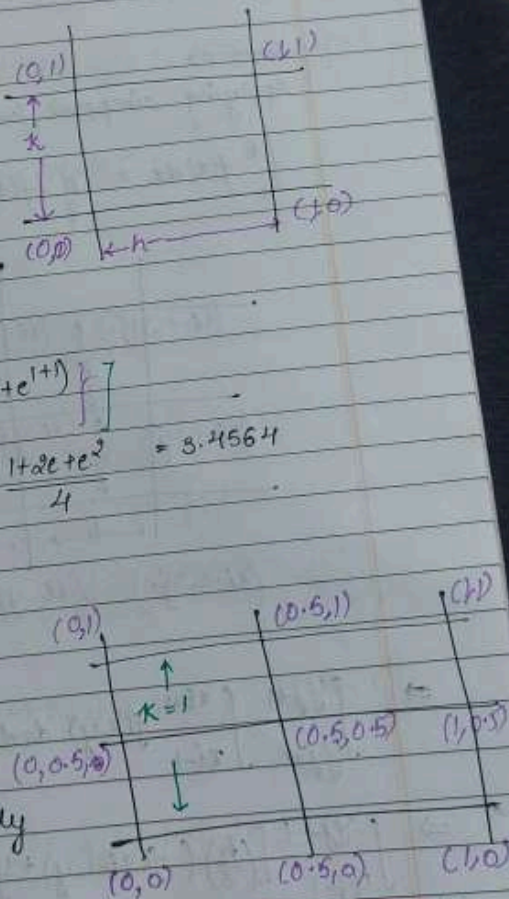
if $(k=1, h=0.5)$:

$$\int_0^1 \int_0^1 e^{x+y} dx dy$$

$$\int_0^1 \left[\left(\frac{0.5}{2} \right) (e^{0+y} + e^{0.5+y} + e^{1+y}) \right] dy$$

$$= \left(\frac{0.5}{2} \right) \left[\left(\int_0^1 e^y dy \right) + \left(\int_0^1 e^{0.5+y} dy \right) + \left(\int_0^1 e^{1+y} dy \right) \right]$$

$$= \left(\frac{0.5}{2} \right) \left[\left(\frac{1-0}{2} \right) (e^0 + e^1) + \left(\frac{1-0}{2} \right) (e^{0.5+0} + e^{0.5+1}) + e^{1+0} + e^{1+1} \right]$$



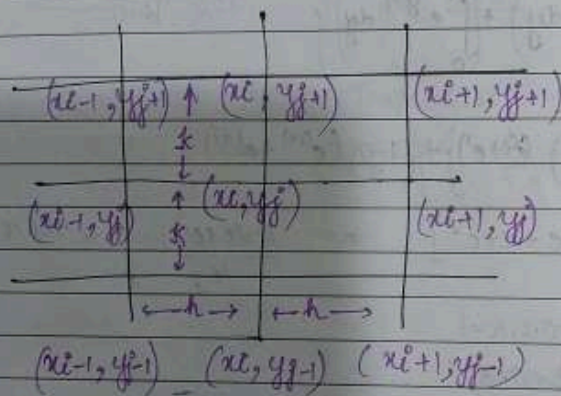
$$= \frac{1}{4} \cdot \frac{1}{2} \{ 1 + e^{\frac{2}{3} \cdot 0.5} + e^{\frac{1}{3} \cdot 0.5} + e^{\frac{2}{3} \cdot 0.5} \}$$

$$= 3.260805$$

Simpson's

Applying Simpson's formula:

$$\int_a^b f(x) dx = \frac{h}{3} \{ f(a) + f(b) + f\left(\frac{a+b}{2}\right) \}$$



$$\Rightarrow \int_{y_{j-1}}^{y_{j+1}} \int_{x_{i-1}}^{x_{i+1}} f(x, y) dx dy$$

$$\Rightarrow \int_{y_{j-1}}^{y_{j+1}} \left[\left(\frac{h}{3} \right) (f(x_{i-1}, y) + 4f(x_i, y) + f(x_{i+1}, y)) \right] dy$$

$$\Rightarrow \left(\frac{h}{3} \right) \left[\int_{y_{j-1}}^{y_{j+1}} f(x_{i-1}, y) dy + 4 \int_{y_{j-1}}^{y_{j+1}} f(x_i, y) dy + \int_{y_{j-1}}^{y_{j+1}} f(x_{i+1}, y) dy \right]$$

$$\Rightarrow \left(\frac{h}{3} \right) \left[\left(\frac{k}{3} \right) f(x_{i-1}, y_{j-1}) + 4 \left(\frac{k}{3} \right) f(x_i, y_{j-1}) + \left(\frac{k}{3} \right) f(x_{i+1}, y_{j-1}) \right]$$

eg $\int_0^1 \int_0^1 e^{x+y} dx dy$

$$= \int_0^1 \left(\frac{h}{3} \right) (e^{0+y} + 4e^{0.5+y} + e^{1+y}) dy$$

$$= \int_0^1 \left(\frac{0.5}{3} \right) (e^y + 4e^{0.5+y} + e^{1+y}) dy$$

$$= \frac{1}{6} \left[\int_0^1 e^y dy + 4 \int_0^1 e^{0.5+y} dy + \int_0^1 e^{1+y} dy \right]$$

$$= \frac{1}{6} \left[\left(\frac{0.5}{3} \right) (e^0 + e^1) + 4 \left(\frac{0.5}{3} \right) (e^{0.5} + e^{1.5}) + \left(\frac{0.5}{3} \right) (e^1 + e^2) \right]$$

$$= \frac{1}{6} \left[1 + 4e^{0.5} + e^1 \right]$$

$$= 2.0483897$$

$$\Rightarrow \left(\frac{h}{3}\right) \left\{ \left(\frac{k}{3}\right) f(x_{i-1}, y_{j-1}) + 4f(x_{i-1}, y_j) + f(x_{i-1}, y_{j+1}) \right. \\ \left. + 4\left(\frac{k}{3}\right) f(x_i, y_{j-1}) + 4f(x_i, y_j) + f(x_i, y_{j+1}) \right. \\ \left. + \left(\frac{k}{3}\right) f(x_{i+1}, y_{j-1}) + 4f(x_{i+1}, y_j) + f(x_{i+1}, y_{j+1}) \right\}$$

eg. $\int_0^1 \int_0^1 e^{x+y} dx dy$; $h=0.5, k=0.5$

$$= \int_0^1 \left(\frac{h}{3}\right) (e^{0+y} + 4e^{0.5+y} + e^{1+y}) dy$$

$$= \int_0^1 \left(\frac{0.5}{3}\right) (e^y + 4e^{0.5+y} + e^{1+y}) dy$$

$$= \frac{1}{6} \left[\int_0^1 e^y dy + 4 \int_0^1 e^{0.5+y} dy + \int_0^1 e^{1+y} dy \right]$$

$$= \frac{1}{6} \left[\left(\frac{0.5}{3}\right) (e^0 + e^{0.5} \times 4 + e^1) + \left(\frac{0.5 \times 4}{3}\right) (e^{0.5+0} + 4e^{0.5+0.5} + e^{0.5+1}) \right. \\ \left. + \left(\frac{0.5}{3}\right) (e^{1+0} + e^{1+0.5} \times 4 + e^{1+1}) \right]$$

$$= \frac{1}{6} \left[1 + 4e^{0.5} + e + 4(e^{0.5} + 4e^1 + e^{1.5}) + e^1 + 4e^{1.5} + e^2 \right]$$

$$= 2.0483897 \quad 2.954483659$$

Numerical solutions of Ordinary Differential Equations

$$\Rightarrow \frac{dy}{dx} = 2y ; y(0) = 5$$

$$\int \frac{dy}{y} = \int 2dx$$

$$\ln|y| = \frac{2x^0}{1} + c \Rightarrow y = e^{2x}c$$

$$\text{at } x=0, y=5$$

$$5=c$$

$$\text{Hence } y = 5e^{2x}$$

$$\Rightarrow \frac{dy}{dx} = \sin(2+y^2), y(0)=1$$

$$y(3) = ?$$

Division into subdivisions/subintervals

$$\begin{array}{ccccccc} y=1 & & & & & & y=2 \\ x=0 & x=1 & x=2 & x=3 \end{array}$$

$$\Rightarrow \frac{dy}{dx} = f(x,y) ; y(x_0) = y_0 \quad (\text{Initial value problem})$$

Integrating it in $[x_n, x_{n+1}]$,

$$\begin{array}{ccccccc} | & | & | & | & \dots & | & | & | \\ x_0 & x_1 & x_2 & x_3 & & x_{n-1} & x_n & x_{n+1} \end{array}$$

$$x_{n+1} \quad x_{n+1}$$

$$\int_{x_n}^{x_{n+1}} dy = \int_{x_n}^{x_{n+1}} f(x,y) dx$$

$$\left[y(x) \right]_{x=x_n}^{x=x_{n+1}} = \int_{x_n}^{x_{n+1}} f(x,y) dx \quad \text{--- (A)}$$

$\frac{dy}{dx}$ is the slope



(a) If x_n and x_{n+1} too much in

from (A),

$$y(x)$$

$$y(x_{n+1})$$

$$y(x)$$

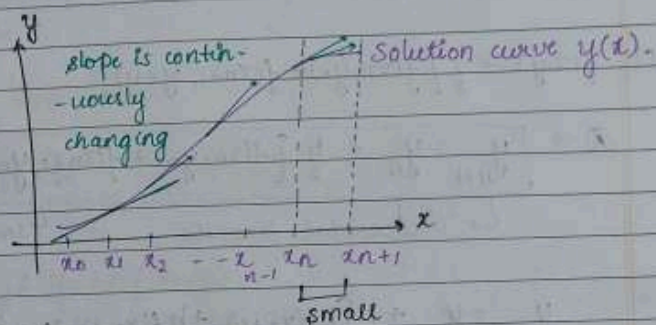
Forward

Euler's formula

$$(b) f(x,y)$$

Now, A $\Rightarrow$

$\frac{dy}{dx}$ is the slope of the tangent to the solution curve y .



- (a) If x_n and x_{n+1} are very close, then, $f(x, y)$ is not changing too much in interval $[x_n, x_{n+1}]$
 $f(x, y) \approx f(x_n, y_n)$

From (a),

$$y(x_{n+1}) - y(x_n) = \int_{x_n}^{x_{n+1}} f(x, y) dx$$

$$y(x_{n+1}) - y(x_n) = f(x_n, y_n) \int_{x_n}^{x_{n+1}} dx$$

$$y(x_{n+1}) = y(x_n) + f(x_n, y_n) (x_{n+1} - x_n)$$

Forward Euler's formula

$$y_{n+1} = y_n + h f(x_n, y_n)$$

Explicit formula (i)

Let $x_i = x_0 + ih$
 $i = 0, 1, 2, \dots$

- (b) $f(x, y) \approx f(x_{n+1}, y_{n+1})$ (Approximation)

Now, A $\Rightarrow$ $y(x_{n+1}) - y(x_n) = f(x_{n+1}, y_{n+1}) \int_{x_n}^{x_{n+1}} dx$

implicit formula

$$y_{n+1} = y_n + h f(x_{n+1}, y_{n+1}) \rightarrow \text{Euler's backward formula}$$

$$(C) f(x, y) = \frac{1}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})]$$

$$(A) \rightarrow y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1})]$$

↳ Trapezoidal formula ↳ implicit

$$y_{n+1} = y_n + \frac{1}{2} [h f(x_n, y_n) + h f(x_{n+1}, y_n + h f(x_n, y_n))]$$

$$y_{n+1} = y_n + \frac{1}{2} (K_1 + K_2) \quad \text{Runge Kutta's Method of second order}$$

$$\text{Here } K_1 = h f(x_n, y_n)$$

$$K_2 = h f(x_{n+1}, y_n + K_1)$$

or

Heun's method

or

Modified Euler's formula.

Ques. Obtain approximated value of $y(1.3)$ for IVP (Initial Value problem) $\frac{dy}{dx} = -2xy^2$, $y(1) = 1$, using Euler's formula

Also, compare it with exact solution.

$$y_0 = 1$$

$$x_0 = 1$$

$$x = h$$

$$y_1 = 2$$

$$x_1 = 1.3$$

By Euler's formula:-

$$y_{n+1} = y_n + h f(x_n, y_n), \quad n = 0, 1, 2, \dots$$

[n=0]

$$y_1 = y_0 + h f(x_0, y_0)$$

$$= 1 + (1.3 - 1) (-2xy^2)$$

$$= 1 + 0.6 \times 1 \times 1^2 \times -1 = 0.4$$

for $h = 0.1$

$$y_0 = 1$$

$$x_0 = 1$$

$$y_{n+1} = \dots$$

$$n=0 \quad y_1 = y_0 + \dots = 1 + \dots$$

$$n=1 \quad y_2 = y_1 + \dots = 0.8 + \dots$$

$$n=3 \quad y_3 = y_2 + \dots = 0.6 + \dots$$

$$= 0.1$$

Exact:-

$$\frac{dy}{dx} = -2xy^2$$

$$\int y^{-2} dy = \int -2x dx$$

$$-\frac{1}{y} = -\frac{2x^2}{2} + \dots$$

$$x^2 + C = 1$$

it formula

y_{n+1} → Euler's backward formula

y_{n+1}

$y_n + f(x_{n+1}, y_{n+1})$

simplicib

$+ h f(x_{n+1}, y_n + h f(x_n, y_n))$

Runge Kutta's Method of second order

$y_n + K_1$ Heun's method

Modified Euler's formula

for IVP (Initial Value Problem) using Euler's formula

solution

$y_1 = 2$

$x_1 = 1.3$

y

y_n , $n=0, 1, 2, \dots$

$2xy^2$
 $-1 = 0.4$

for $h=0.1$

$y_0=1$ $x_0=1$ y_1 $x_1=1.1$ y_2 $x_2=1.2$ y_3 $x_3=1.3$

$y_{n+1} = y_n + h f(x_n, y_n)$; $n=0, 1, 2, \dots$

$$n=0 \quad y_1 = y_0 + h f(x_0, y_0) = 1 + (0.1)(-2 \times 1 \times 1^2) = 1 - 0.2 = 0.8$$

$$n=1 \quad y_2 = y_1 + h f(x_1, y_1) = 0.8 + (0.1)(-2 \times 1.1 \times 0.8^2) = 0.8(1 - 2 \times 0.1 \times 1.1 \times 0.8) = 0.8 - 0.2368 = +0.5632$$

$$n=2 \quad y_3 = y_2 + h f(x_2, y_2) = 0.5632 + (0.1)(-2 \times 1.2 \times 0.5632^2) = 0.5549$$

Exact:-

$$\frac{dy}{dx} = -2xy^2$$

$$\int y^{-2} dy = \int -2x dx$$

$$-\frac{1}{y} = -\frac{2x^2}{2} + C$$

$$x^2 + C = \frac{1}{y}$$

$$y(1) = 1$$

$$1^2 + C = 1$$

$$C = 0$$

$$\text{so, } y = \frac{1}{x^2}$$

$$x=1.3 \quad y=0.5917$$